

ERP HRMS Technical Documentation

Employee Performance Management and Learning Management Modules

Integrated with Recruitment, Onboarding, Core HR, Workforce Management, Succession Planning, Workflow, Audit, Reporting, IAM, and Finance

Document Field	Value
Document Type	Technical Implementation Documentation
Audience	Developers, Solution Architects, Infrastructure Team, DevOps, QA, HR Product Owners, Security, Data Team
Scope	Performance Management and Learning Management modules integrated into the existing ERP HRMS design
Design Style	Enterprise ERP / SAP-style HCM architecture, implementation-agnostic
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Implementation principle: This document defines what the dev and infra teams need to build and configure without locking the solution into SAP, Django, Java, .NET, Node.js, or any single vendor. The design can be implemented as a modular monolith, service-oriented architecture, or microservices architecture depending on enterprise maturity.

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1. Executive Summary

This technical documentation extends the current ERP HRMS design with two additional enterprise modules: Employee Performance Management and Learning Management. These modules must integrate with the existing Recruitment, Onboarding, Core HR, Workforce Management, Succession Planning, Workflow, Audit, Reporting, IAM, and Finance capabilities.

- Performance Management will manage performance cycles, goals, competencies, check-ins, self-reviews, manager reviews, 360 feedback, calibration, final ratings, performance improvement plans, and development recommendations.
- Learning Management will manage course catalog, learning paths, mandatory training, training requests, enrollments, classroom/online sessions, assessments, certifications, learning transcript, and development-plan-linked learning.
- Recruitment and Onboarding must feed new-hire role expectations, skills, probation goals, and induction learning requirements into the new modules.
- Succession Planning must consume performance outcomes, learning completions, certifications, competency evidence, and development plan progress.
- The design should support future AI capabilities such as skills inference, recommended learning, goal generation, talent marketplace, and succession readiness prediction.

Reference model: The design follows common SAP-style HCM concepts: integrated talent management, performance and goals, continuous feedback, learning, compliance training, skills management, succession, and development. It is not a SAP implementation document and should remain vendor-neutral.

2. System Context and Existing HRMS Baseline

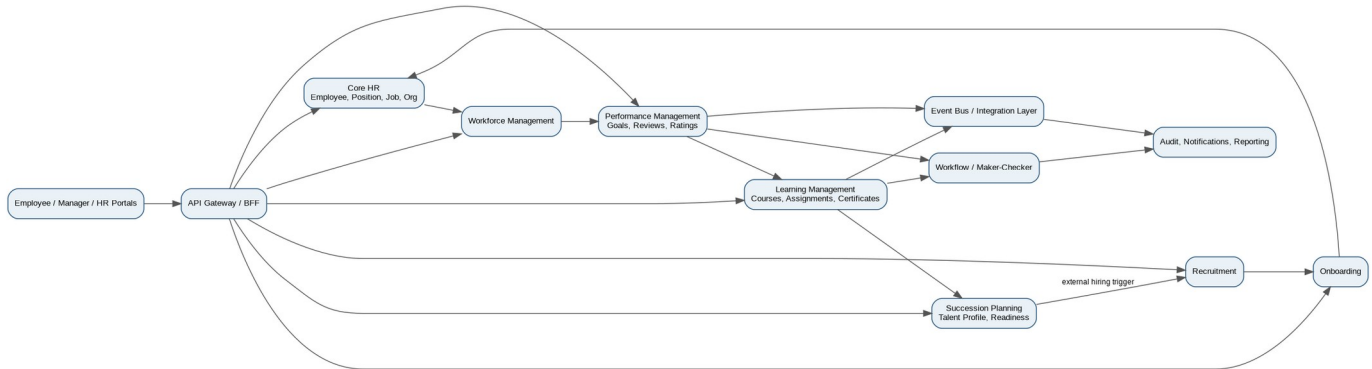
The existing ERP HRMS baseline already includes Core HR, Recruitment, Onboarding, Workforce Management, Succession Planning, maker-checker workflow, audit, reporting, and integrations. The new modules must connect into that same enterprise backbone instead of creating duplicate employee, position, job, department, role, or approval tables.

2.1 Existing HRMS Modules

Existing Module	Existing Responsibility	Dependency for New Modules
Core HR	Employee master, person data, employment data, organization structure, position, job, grade, manager assignment	Primary master data source for performance and learning eligibility
Recruitment	Requisition, job posting, applicant portal, candidate pipeline, interview, offer	Feeds new-hire job expectations, required skills, candidate skills, and probation setup
Onboarding	Checklist, document collection, access request, asset request, employee conversion	Triggers induction learning and probation performance goals
Workforce Management	Position control, assignment, schedule, attendance, overtime, manpower planning	Provides current assignment, department, manager, job, location, and work context
Succession Planning	Critical role register, successor nomination, talent profile, readiness, development plan	Consumes ratings, competency evidence, certifications, learning completion, and development progress
Workflow	Maker-checker approval, approval routing, comments, SLA, escalation	Required for rating approvals, course publishing, external training approvals, waiver approvals
Audit	Transaction log, before/after values, approval log	Required for performance and learning governance
Reporting	Executive dashboards and operational reports	Adds performance, learning, skills, compliance, and readiness dashboards

3. Target Architecture

The architecture below extends the HRMS with Performance and Learning as two domain modules connected through the same API gateway, workflow engine, shared master data, event bus, audit layer, and reporting platform.



Architecture Layer	Recommended Components	Responsibilities
Presentation Layer	Employee portal, Manager portal, HR admin portal, Executive dashboard, Mobile-responsive UI	Self-service, manager reviews, HR configuration, dashboards
API Layer	API Gateway, Backend-for-Frontend, REST/GraphQL APIs	Authentication, authorization, routing, request validation, API versioning
Domain Layer	Performance service/module, Learning service/module, Core HR service/module, Workflow service/module	Business rules, state transitions, maker-checker, domain transactions
Integration Layer	Event bus, message queue, integration adapters, scheduled jobs	Cross-module events, async processing, external LMS/content provider integration
Data Layer	PostgreSQL/RDBMS, Redis/cache, object storage, search index, reporting mart	Master data, transactions, files/certificates, analytics
Security Layer	IAM/SSO, RBAC, SoD policies, secrets management, encryption	Identity, permissions, access control, data protection
Operations Layer	CI/CD, Kubernetes/OpenShift, monitoring, logging, backup, DR	Deployment, reliability, observability, restore capability

3.1 Integrated Talent Lifecycle



3.2 Recommended Deployment Options

Option	Description	When to Use
Modular monolith	One deployable application with separated modules/packages and shared database schema	Fast POC to MVP, smaller team, easier transactions
Service-oriented architecture	Core HR, Workflow, Performance, Learning, Recruitment and Succession are separate services/modules with shared integration contracts	Enterprise rollout, multiple teams, clear module ownership
Microservices	Fully independent services with separate databases and event-driven integration	Mature DevOps, strong platform engineering, high scale, strict service boundaries

4. Module Boundaries and Service Design

The new modules must be designed with clear boundaries. They should reuse Core HR master data and Workflow instead of duplicating employee, position, organization, manager, approval, and audit logic.

Module / Service	Owns	Consumes	Publishes
Performance Management	Cycles, templates, goals, check-ins, reviews, ratings, calibration, PIP	Employee, position, job, org, competency model, manager assignment, learning evidence	Performance result, competency gap, PIP status, talent profile update event
Learning Management	Course catalog, learning paths, assignments, enrollments, completions, certificates, transcripts	Employee, position, job, org, performance gaps, succession development plan, budget data	Learning completion, certification status, compliance risk, skill update event
Core HR	Employee, position, job, grade, department, manager assignment	Approved onboarding conversion, employee lifecycle changes	Employee created, employee assigned, manager changed, position changed
Recruitment	Requisitions, jobs, applicants, offers	Approved position, succession hiring trigger	Candidate hired, offer accepted, candidate skills
Onboarding	Onboarding checklist, documents, access/asset tasks	Offer accepted, employee master draft	Onboarding completed, new hire learning required
Workflow	Approval requests, steps, comments, status, SLA	Workflow configuration from modules	Approval completed, rejected, returned, escalated
Succession	Critical roles, successor nomination, readiness, talent profile	Performance rating, learning evidence, skills, certifications	Development plan created, external recruitment trigger

Boundary rule: Performance and Learning must not store their own employee master copies. They may store employee_id as a foreign key/reference and a point-in-time snapshot only when required for historical accuracy.

5. Employee Performance Management Technical Specification

The Performance Management module manages performance planning, continuous feedback, formal review, calibration, final rating, improvement plans, and downstream integration to learning, succession, and compensation.



5.1 Functional Components

Component	Purpose	Key Capabilities
Performance Cycle Manager	Controls review period and eligibility	Cycle dates, eligibility population, review templates, reminders, lock dates
Goal Management	Creates and tracks goals	Goal library, cascading goals, individual goals, weights, milestones, progress updates
Competency Assessment	Assesses skills and behaviors	Competency model, role-based competency template, manager rating, evidence
Continuous Feedback	Captures ongoing feedback	Check-ins, 1:1 notes, achievements, coaching notes, manager comments
Review Forms	Formal self and manager review	Self-review, manager review, comments, ratings, attachments
360 Feedback	Optional multi-rater feedback	Peer, matrix manager, stakeholder, subordinate feedback
Calibration	Normalizes ratings	Pre-calibration rating, calibrated rating, distribution rules, committee decision
Final Outcome	Stores approved rating	Final rating, effective date, release status, employee acknowledgment
PIP	Manages underperformance	Improvement plan, milestones, check-ins, closure, escalation
Integration Adapter	Publishes outcomes	Learning gap event, succession update, compensation input, reporting mart update

5.2 Performance Cycle Configuration

Configuration Item	Description	Example
Cycle type	Defines whether annual, mid-year, probation, project-based, or ad hoc	Annual 2026, Probation Review
Eligibility rule	Determines employees included in the cycle	Active employees hired before cutoff date
Review template	Defines sections and weightings	Goals 60%, Competencies 30%, Values 10%
Rating scale	Scale and labels used for scoring	1 to 5 rating scale
Workflow route	Approval chain for final review	Manager -> HR Checker -> Department Head
Lock dates	Dates after which goals or ratings cannot be edited without approval	Goal lock date, review submission deadline
Notification schedule	Reminder rules	14, 7, 3 days before deadline
Calibration required	Whether calibration is mandatory before final approval	Required for annual reviews

5.3 Performance Workflow Rules

Workflow	Trigger	Maker	Checker / Approver	Outcome
Cycle publishing	HR creates or updates cycle	HR Performance Officer	HR Performance Checker / HR Head	Cycle becomes active
Goal approval	Employee or manager submits goals	Employee / Manager	Manager / HR if disputed	Goals locked for cycle
Manager review	Employee self-review submitted	Manager	HR Checker	Proposed rating reviewed
Calibration	Review period reaches calibration stage	Department Head / Committee	HR Checker / HR Head	Calibrated rating stored
Final release	Ratings approved	HR Performance Officer	HR Checker	Results visible to employee
Rating override	Manager or HR requests change	Manager / HR	HR Head	New final rating with audit reason
PIP creation	Low rating or manager escalation	Manager	HR Checker / Employee Relations	PIP activated

5.4 Performance Status Model

NOT_STARTED
 GOAL_SETTING
 GOALS_SUBMITTED
 GOALS_LOCKED
 IN_PROGRESS
 SELF_REVIEW_PENDING
 MANAGER_REVIEW_PENDING
 CALIBRATION_PENDING
 CHECKER_REVIEW_PENDING
 RETURNED_FOR_AMENDMENT
 FINAL_APPROVED
 RELEASED_TO_EMPLOYEE
 EMPLOYEE_ACKNOWLEDGED
 CLOSED
 CANCELLED

5.5 Key Validations

- Only active employees eligible under the cycle rule may receive review forms.
- A goal must have title, type, weight, due date, status, and owner before submission.
- Total goal weighting must equal configured total, for example 100%.
- Goal changes after lock date require workflow approval and audit reason.
- Manager cannot final approve their own rating unless explicitly allowed by configuration.
- Final rating must not be released until checker approval and calibration completion, if configured.
- Rating override must store previous rating, new rating, reason, approver, timestamp, and evidence.
- PIP cannot be closed without outcome, evidence, manager comment, and HR checker approval.
- Final performance outcome must update talent profile and reporting mart.

5.6 Performance API Endpoints

Method	Endpoint	Purpose	Main Roles
POST	/api/performance/cycles	Create performance cycle draft	HR Performance Maker
PATCH	/api/performance/cycles/{id}	Update cycle draft	HR Performance Maker
POST	/api/performance/cycles/{id}/submit	Submit cycle for approval	HR Performance Maker
POST	/api/performance/cycles/{id}/publish	Publish approved cycle	HR Checker / HR Head
GET	/api/performance/cycles/{id}/eligible-employees	Preview cycle population	HR Performance / HR Checker
POST	/api/performance/goal-plans	Create employee goal plan	Manager / Employee
POST	/api/performance/goals/{id}/submit	Submit goal for approval	Employee / Manager
PATCH	/api/performance/goals/{id}/progress	Update goal progress	Employee / Manager
POST	/api/performance/reviews/{id}/self-review	Submit self assessment	Employee
POST	/api/performance/reviews/{id}/manager-review	Submit manager assessment	Manager
POST	/api/performance/reviews/{id}/calibrate	Submit calibrated rating	Department Head / HR
POST	/api/performance/reviews/{id}/approve	Approve final rating	HR Checker / Approver
POST	/api/performance/pips	Create PIP	Manager / HR
GET	/api/performance/reports/cycle-progress	Cycle progress report	HR / Executive

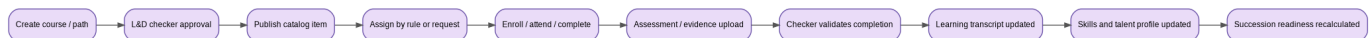
5.7 Example Performance Review Payload

Sample JSON - performance review result

```
{
  "review_id": "PERF-REV-2026-000123",
  "cycle_id": "PERF-CYCLE-2026-ANNUAL",
  "employee_id": "EMP-000987",
  "manager_id": "EMP-000111",
  "review_sections": [
    {"type": "GOALS", "weight": 60, "score": 4.1},
    {"type": "COMPETENCY", "weight": 30, "score": 3.8},
    {"type": "VALUES", "weight": 10, "score": 4.0}
  ],
  "proposed_rating": 4,
  "calibrated_rating": 4,
  "final_rating": 4,
  "status": "FINAL_APPROVED",
  "skill_gaps": ["leadership_communication", "data_storytelling"],
  "learning_recommendation_required": true
}
```

6. Learning Management Technical Specification

The Learning Management module manages training catalog, learning paths, mandatory assignments, enrollments, completion evidence, certifications, learning transcripts, training requests, external training approvals, skills gaps, and development-plan-linked learning.



6.1 Functional Components

Component	Purpose	Key Capabilities
Course Catalog	Central learning item repository	Course metadata, category, provider, format, duration, prerequisites
Learning Path Manager	Groups courses into curriculum	Role-based paths, onboarding paths, leadership paths, compliance paths
Assignment Engine	Assigns learning to employees	Rules by job, position, grade, department, location, compliance need, performance gap
Enrollment Manager	Handles self-enrollment and approvals	Waitlist, session capacity, approvals,

		cancellation
Session Manager	Manages classroom/live sessions	Instructor, venue, calendar, attendance, capacity
Assessment Engine	Manages tests and scoring	Quiz, pass mark, attempts, retake rules, completion score
Completion Validator	Validates learning completion	Automatic completion, manual validation, certificate evidence
Certification Manager	Tracks certification requirement and expiry	Issue date, expiry date, renewal reminders, evidence
Learning Transcript	Stores learner history	Completed courses, scores, certificates, external training
Training Budget	Tracks external training cost	Budget check, finance approval, actual cost
Integration Adapter	Publishes learning outcomes	Skill updates, compliance status, succession evidence, performance evidence

6.2 Learning Assignment Sources

Source	Assignment Logic	Example
Onboarding	Assign when onboarding starts or employee record is activated	Corporate induction, security awareness, HR policies
Job / Position	Assign based on current job or position requirement	Relationship manager certification, branch operations training
Compliance	Assign based on regulation, department, role, or risk classification	AML, data protection, ethics, cyber security
Performance	Assign based on skill or competency gap	Communication workshop after competency gap
Succession	Assign based on development plan for successor readiness	Leadership program, acting manager training
Employee Request	Employee requests training from catalog	External conference or certification request
Manager Assignment	Manager assigns learning to team member	New process training for team

6.3 Learning Workflow Rules

Workflow	Trigger	Maker	Checker / Approver	Outcome
Course creation	L&D creates course	L&D Officer	L&D Checker	Course approved for publishing
Mandatory learning rule	Compliance training created	L&D / Compliance	HR Compliance Checker	Rule assigned to target population
External training request	Employee submits training request	Employee / Manager	Manager + L&D + Finance	External training approved or rejected
Certificate validation	Employee uploads certificate	Employee	L&D Checker / Compliance	Certificate marked valid or returned
Manual completion	Trainer marks employee complete	Trainer / L&D	L&D Checker	Learning transcript updated
Waiver request	Employee or manager requests waiver	Manager / L&D	Compliance Head / HR Head	Mandatory training waived with audit reason
Budget approval	Training has cost impact	Manager / Employee	Finance Checker	Training cost reserved or rejected

6.4 Learning Status Model

ASSIGNED
 ENROLLED
 WAITLISTED
 IN_PROGRESS
 COMPLETED_PENDING_VALIDATION
 COMPLETED_VERIFIED
 FAILED
 CANCELLED
 WAIVED
 OVERDUE
 EXPIRED
 RENEWAL_REQUIRED

6.5 Key Validations

- A course cannot be published until required metadata, owner, target audience, delivery mode, and approval status are complete.
- Mandatory learning must define target population, due date, escalation rule, and owner.
- External training with cost must pass manager, L&D, and finance approval based on policy.
- A certificate must include issuer, issue date, expiry date when applicable, evidence attachment, and validation status.
- Manual completion must require checker validation to prevent false completion records.
- Waiver of mandatory learning must require senior approval and cannot remove audit history.
- Learning completion must update the learning transcript and publish a learning.completed event.
- Expired certification must trigger notification and compliance risk flag.

6.6 Learning API Endpoints

Method	Endpoint	Purpose	Main Roles
POST	/api/learning/courses	Create course draft	L&D Maker
PATCH	/api/learning/courses/{id}	Update course draft	L&D Maker
POST	/api/learning/courses/{id}/submit	Submit course for approval	L&D Maker
POST	/api/learning/courses/{id}/publish	Publish approved course	L&D Checker
POST	/api/learning/paths	Create learning path	L&D Maker
POST	/api/learning/assignment-rules	Create assignment rule	L&D / Compliance
POST	/api/learning/assignments	Assign learning manually	Manager / L&D
POST	/api/learning/enrollments	Enroll in course or session	Employee / Manager
POST	/api/learning/completions	Submit completion evidence	Employee / Trainer
POST	/api/learning/completions/{id}/validate	Validate completion	L&D Checker
POST	/api/learning/certificates	Upload certificate evidence	Employee
POST	/api/learning/training-requests	Request internal/external training	Employee / Manager
GET	/api/learning/transcript/{employee_id}	View learning transcript	Employee / Manager / HR
GET	/api/learning/reports/compliance	Mandatory training compliance report	HR / Compliance

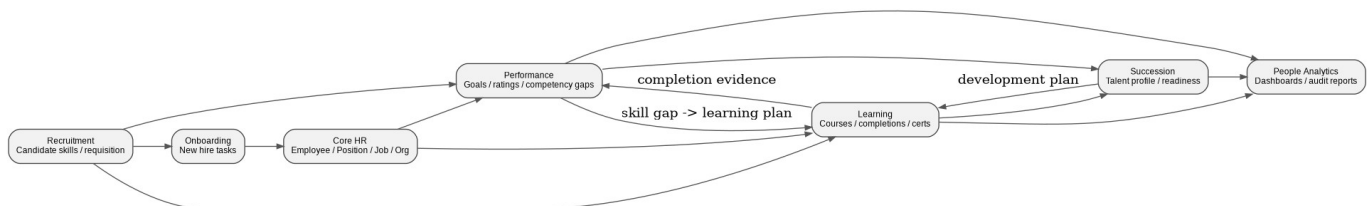
6.7 Example Learning Assignment Payload

Sample JSON - learning assignment

```
{
  "assignment_id": "LRN-ASG-2026-000455",
  "employee_id": "EMP-000987",
  "source_module": "PERFORMANCE",
  "source_reference_id": "PERF-REV-2026-000123",
  "learning_item_type": "LEARNING_PATH",
  "learning_item_id": "LP-LEADERSHIP-FOUNDATION",
  "assignment_reason": "Skill gap identified during annual review",
  "mandatory": false,
  "due_date": "2026-09-30",
  "status": "ASSIGNED",
  "requires_validation": true
}
```

7. Integration with Recruitment, Onboarding, Workforce, and Succession

The new modules must be wired into the existing HRMS using shared IDs, events, APIs, and workflow. The diagram below shows the integrated talent data flow.



7.1 Recruitment Integration Requirements

Recruitment Data	Performance Usage	Learning Usage	Technical Requirement
Job requisition	Creates initial role expectation and probation goal template	Determines role-based induction learning	Requisition must reference position_id and job_id
Job description	Provides goal and competency template baseline	Maps required skill to learning path	Maintain structured job requirements, not free text only
Candidate skills	Initial skills profile for first review	Skills gap and learning recommendation	Capture normalized skill tags during recruitment
Offer grade	Determines eligible review template and rating rules	Determines learning eligibility and mandatory training	Offer must include grade_id and employment type
Hiring manager	Becomes first performance reviewer	Approves early training requests	Map hiring manager to manager_id after hire
Candidate documents	Supports qualification evidence	Supports certificate validation	Document metadata must be carried into HR document store

7.2 Onboarding Integration Requirements

- When offer is accepted, onboarding must create a new hire onboarding case.
- When onboarding starts, Learning must assign induction and compliance learning based on job, department, location, employment type, and policy.
- When employee master is activated, Performance must create probation goals if the employee is eligible for probation review.
- When onboarding is complete, employee.learning_profile_initialized and employee.performance_profile_initialized events should be published.
- Onboarding checklist completion should include training prerequisites such as code of conduct, information security, and role-specific training.

7.3 Workforce Integration Requirements

Workforce Data	Performance Impact	Learning Impact
Current position	Determines goal/competency template	Determines learning path and certification requirements
Department	Determines department goals and calibration group	Determines department mandatory training
Manager assignment	Determines reviewer and approver	Determines training request approver
Location / branch	Can affect operational KPIs	Can assign location-specific training
Grade	Determines review form and rating policy	Determines eligible courses/programs
Attendance / leave context	Optional evidence during review	Helps schedule training capacity

7.4 Succession Integration Requirements

- Final performance outcome must update the employee talent profile with rating, competency score, comments summary, and review cycle metadata.
- Learning completion must update readiness evidence for successor candidates.
- Certification records must be available to Succession Planning for critical roles that require regulatory, technical, or leadership certification.
- Succession development plans must create linked learning assignments in the Learning module.
- When no successor is ready, Succession may trigger Recruitment. Performance and Learning history should be visible to HR Talent for internal candidate evaluation.

8. Workflow and Maker-Checker Technical Design

The workflow engine should be generic and reusable across HRMS modules. Performance and Learning must create workflow requests instead of hardcoding approval logic into each screen.

8.1 Generic Workflow Model

Entity	Purpose	Important Fields
workflow_request	Parent workflow record for any approval	id, module, entity_type, entity_id, status, maker_id, current_step, created_at, submitted_at
workflow_approval_step	Each approval stage	id, workflow_id, step_no, approver_role, approver_id, status, action_at, due_at
workflow_comment	Approval comments and return reasons	id, workflow_id, step_id, user_id, comment, visibility, created_at
workflow_attachment	Supporting evidence	id, workflow_id, file_id, attachment_type, uploaded_by
workflow_rule	Approval routing configuration	module, process_code, condition_expression, approver_role, SLA

8.2 Maker-Checker Matrix

Process	Maker	Checker	Final Approver	Mandatory Audit Reason?
Performance cycle creation	HR Performance Officer	HR Checker	HR Head	Yes for rejection/return
Review template publication	HR Performance Officer	HR Checker	HR Head	Yes
Final rating approval	Manager	HR Checker	Department Head / HR Head	Yes for override
Calibration adjustment	Department Head / Committee	HR Checker	HR Head	Yes
PIP creation	Manager	HR Checker	Employee Relations	Yes
Course creation	L&D Officer	L&D Checker	Optional HR Head	Yes for rejection/return
Mandatory training rule	L&D / Compliance	HR Compliance Checker	Compliance Head	Yes
External training approval	Employee / Manager	Manager + L&D + Finance	Department Head if above threshold	Yes
Certificate validation	Employee	L&D Checker	Compliance if regulated	Yes for rejection
Mandatory training waiver	Manager / L&D	HR Checker	Compliance Head / HR Head	Yes

8.3 Workflow State Transitions

DRAFT -> SUBMITTED -> IN_REVIEW -> RETURNED -> RESUBMITTED -> APPROVED -> ACTIVE

DRAFT -> SUBMITTED -> IN_REVIEW -> REJECTED

ACTIVE -> AMENDMENT_REQUESTED -> SUBMITTED -> APPROVED -> ACTIVE

ACTIVE -> CANCELLED

9. Data Model and Key Tables

The following tables are implementation-level guidance. Your team may rename tables based on internal standards, but the entity ownership, relationships, and audit/effective dating requirements should remain.

9.1 Performance Tables

Table	Purpose	Core Fields
perf_cycle	Performance cycle master	cycle_id, name, type, start_date, end_date, status, eligibility_rule_id, template_id
perf_cycle_template	Review form template	template_id, name, sections_json, rating_scale_id, active_flag
perf_goal_plan	Employee goal plan per cycle	goal_plan_id, cycle_id, employee_id, manager_id, status, total_weight
perf_goal	Individual goal	goal_id, goal_plan_id, title, description, type, weight, target_value, due_date, status
perf_goal_progress	Goal updates	progress_id, goal_id, progress_percent, comment, updated_by, updated_at
perf_competency_model	Role competency model	model_id, job_id, position_id, grade_id, effective_date
perf_competency_item	Competency details	competency_id, model_id, name, level_required, weight
perf_review_form	Employee review form	review_id, cycle_id, employee_id, manager_id, status
perf_review_rating	Scores by section	rating_id, review_id, section_code, score, weight, weighted_score
perf_review_comment	Review comments	comment_id, review_id, author_id, visibility, comment_text
perf_calibration_session	Calibration meeting	session_id, cycle_id, department_id, owner_id, status
perf_calibrated_rating	Calibrated result	id, review_id, proposed_rating, calibrated_rating,

		reason, approved_by
perf_final_outcome	Approved result	outcome_id, review_id, final_rating, release_date, acknowledged_at
perf_improvement_plan	PIP master	pip_id, employee_id, manager_id, reason, start_date, end_date, status
perf_improvement_milestone	PIP milestones	milestone_id, pip_id, title, due_date, outcome, evidence_file_id

9.2 Learning Tables

Table	Purpose	Core Fields
learn_course	Course catalog item	course_id, title, description, category_id, provider_id, delivery_mode, duration, status
learn_course_content	Course content metadata	content_id, course_id, content_type, file_id/url, duration, sequence_no
learn_learning_path	Learning curriculum	path_id, name, type, owner_id, status
learn_learning_path_item	Courses in path	path_item_id, path_id, course_id, sequence_no, mandatory_flag
learn_assignment_rule	Auto-assignment rules	rule_id, source, condition_expression, learning_item_id, due_days
learn_assignment	Employee learning assignment	assignment_id, employee_id, item_type, item_id, due_date, status, source_module
learn_enrollment	Course/session enrollment	enrollment_id, assignment_id, session_id, status, enrolled_at
learn_session	Class or virtual session	session_id, course_id, trainer_id, start_time, end_time, venue, capacity
learn_session_attendance	Attendance for session	id, session_id, employee_id, attendance_status, validated_by
learn_assessment	Test/assessment	assessment_id, course_id, pass_score, max_attempts
learn_assessment_result	Assessment result	result_id, assessment_id, employee_id, score, passed, attempt_no
learn_completion	Completion record	completion_id, assignment_id, employee_id, completion_date, validation_status
learn_certificate	Certificate master/evidence	certificate_id, employee_id, issuer, issue_date, expiry_date, file_id, status
learn_training_request	Training request workflow	request_id, employee_id, course_id/external_title, cost, status
learn_learning_transcript	Historical transcript	transcript_id, employee_id, item_id, completion_status, score, certificate_id
learn_skill_gap	Skills gap record	gap_id, employee_id, skill_id, source_module, source_reference_id, status

9.3 Shared Keys and Indexing Guidance

Area	Index / Constraint Guidance
Identity references	Use employee_id, position_id, job_id, department_id, manager_id from Core HR. Enforce referential integrity where technically feasible.
Workflow lookup	Index workflow_request(module, entity_type, entity_id, status).
Performance cycle lookup	Index perf_review_form(cycle_id, employee_id), perf_goal_plan(cycle_id, employee_id).
Learning lookup	Index learn_assignment(employee_id, status, due_date), learn_completion(employee_id, completion_date).
Compliance reporting	Index learn_certificate(employee_id, expiry_date, status), learn_assignment(status, mandatory_flag, due_date).
Auditability	All high-risk tables require created_by, created_at, updated_by, updated_at, version, and soft-delete/status fields.
Historical accuracy	Store snapshot fields for manager, department, position, and grade on review finalization to preserve historical context.

10. API and Event Contracts

APIs support synchronous user actions. Events support cross-module integration and async processing. Use both: do not rely only on direct database reads across module boundaries.

10.1 Recommended Events

Event Name	Publisher	Consumers	Purpose
employee.created	Core HR	Learning, Performance, IAM, Reporting	Initialize new employee profiles
employee.assigned_to_position	Workforce/Core HR	Performance, Learning, Succession	Assign role-based goals and learning
offer.accepted	Recruitment	Onboarding, Learning, Performance	Prepare onboarding and probation setup
onboarding.completed	Onboarding	Core HR, Learning, Performance	Activate employee and confirm induction status
performance.cycle.published	Performance	Notification, Reporting	Notify eligible employees/managers
performance.review.finalized	Performance	Learning, Succession, Compensation, Reporting	Create learning gaps and update talent profile
performance.pip.created	Performance	HR Case, Learning, Reporting	Track underperformance and development action
learning.assignment.created	Learning	Notification, Reporting	Notify learner and manager
learning.completed	Learning	Performance, Succession, Reporting	Update evidence and learning transcript
certification.expiring	Learning	Employee, Manager, Compliance	Notify renewal requirement
succession.development_plan.created	Succession	Learning	Create successor development learning plan

10.2 Event Envelope Standard

Event envelope contract

```
{
  "event_id": "uuid",
  "event_name": "performance.review.finalized",
  "event_version": "1.0",
  "occurred_at": "2026-05-25T10:00:00+08:00",
  "publisher": "performance-service",
  "correlation_id": "request-or-workflow-id",
  "actor_id": "EMP-ADMIN-001",
  "tenant_id": "default",
  "payload": {
    "review_id": "PERF-REV-2026-000123",
    "employee_id": "EMP-000987",
    "cycle_id": "PERF-CYCLE-2026-ANNUAL",
    "final_rating": 4,
    "skill_gaps": ["leadership_communication"],
    "effective_date": "2026-12-31"
  }
}
```

10.3 API Design Standards

- All APIs must require authentication through enterprise SSO or token-based IAM.
- All write APIs must enforce RBAC and maker-checker policy before state change.
- All APIs must return correlation_id for tracing.
- Use optimistic locking with version fields for editable records such as goals, reviews, courses, and learning paths.
- Use pagination, filters, and sorting for list endpoints.
- Do not expose confidential fields such as calibration notes, PIP records, and succession-linked ratings unless role policy allows it.
- Use idempotency keys for workflow submissions and completion updates to avoid duplicate transactions.

11. UI/UX Screens and Navigation

Menu	Screens	Primary Users
Performance Dashboard	Cycle progress, pending reviews, rating distribution, overdue actions	HR, Managers, Executives

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My Performance	My goals, self-review, feedback, review result, acknowledgment	Employee
Team Performance	Team goals, pending self-reviews, manager review forms, ratings	Manager
Performance Admin	Cycle setup, templates, rating scale, eligibility, reminders	HR Performance
Calibration	Department rating grid, distribution, proposed vs calibrated ratings	Department Head, HR
PIP Management	PIP cases, milestones, check-ins, closure evidence	Manager, HR Employee Relations
Learning Catalog	Search courses, learning paths, details, enrollment	Employee, Manager
My Learning	Assignments, due dates, completions, certificates, transcript	Employee
Team Learning	Team learning compliance, overdue items, requests	Manager
Learning Admin	Courses, paths, sessions, assignment rules, mandatory training	L&D
Certification Management	Certificates, validation, expiry dashboard, renewals	Employee, L&D, Compliance
Training Requests	Internal/external training requests, approvals, budget status	Employee, Manager, Finance, L&D

11.1 Screen-Level Build Notes

- Every maker-checker screen must show current status, current approver, submitted date, returned reason, and approval timeline.
- Manager dashboards must prioritize pending actions: goal approval, self-review review, team learning overdue, training request approval.
- HR admin screens must support export, filtering, version history, and audit trail visibility.
- Employee screens must show only personal performance and learning records, not confidential calibration or succession notes.
- Learning catalog should support search by skill, job family, category, delivery mode, mandatory/optional, and certification relevance.
- Performance review screens should autosave drafts and warn before submission.

12. Security, RBAC, Audit, and Compliance

12.1 RBAC Matrix

Capability	Employee	Manager	HR Performance	L&D	HR Checker	Executive
View own review	Yes	No	Yes	No	Yes	Aggregate only
Edit own goals	Yes, before lock	Team only	No	No	No	No
Submit manager rating	No	Team only	No	No	Review only	No
Approve final rating	No	No	No	No	Yes	No
View calibration notes	No	Limited	Yes	No	Yes	Aggregate only
Create course	No	No	No	Yes	Review only	No
Enroll in learning	Own only	Team assign	Yes	Yes	Yes	No
Validate certificate	No	No	No	Yes	Yes	No
Waive mandatory learning	No	Request only	No	Request only	Review	No
View executive dashboard	No	Team only	Yes	Yes	Yes	Yes

12.2 Audit Requirements

- Audit every create, update, submit, approve, reject, return, override, waive, publish, complete, and validate action.
- Store before and after values for ratings, goals, course rules, mandatory assignments, certificate status, and PIP status.
- Store user, role, timestamp, IP/device metadata where available, correlation_id, workflow_id, and reason.
- Audit logs must be immutable to normal application users.

- Provide audit export for internal audit, compliance, and HR governance review.

12.3 Sensitive Data Protection

Data	Protection
Final ratings	Restricted by employee, manager, HR, and authorized approvers
Calibration notes	Restricted to HR and calibration committee
PIP records	Highly restricted HR employee-relations records
360 feedback	Visibility controlled by policy and anonymization configuration
Learning compliance failures	Visible to employee, manager, HR compliance, and authorized L&D
Certificates and documents	Stored in secure object storage with access control and audit
Succession-linked development plans	Visible only to HR Talent and authorized managers

13. Infrastructure and Deployment Configuration

The modules should be deployable into the same enterprise platform as the existing HRMS. The design below is vendor-neutral but assumes standard enterprise practices: containerized workloads, CI/CD, secure secrets, environment separation, centralized logging, monitoring, and backup.

13.1 Runtime Components

Component	Purpose	Configuration Notes
Frontend app	Employee, manager, HR, L&D, executive screens	Can be React/Vue/Angular/server-rendered; connect through API gateway
Performance backend	Performance cycle, goals, reviews, calibration, PIP	Stateless service; scale horizontally
Learning backend	Catalog, assignments, completion, certificates	Stateless service; background jobs for reminders and expiry checks
Workflow backend	Maker-checker approvals	Shared by all HR modules; enforce SoD
PostgreSQL/RDBMS	Transactional database	Enable backups, PITR, indexes, FK constraints, audit schema
Redis/cache	Session, cache, locks, queues if needed	Use for cache and short-lived state only, not system of record
Message broker/event bus	Async integration events	Kafka/RabbitMQ/Azure Service Bus/etc.
Object storage	Certificates, course content, evidence files	Use encryption and access-controlled URLs
Search index	Course catalog and employee/talent search	Optional OpenSearch/Elasticsearch or DB full-text
Reporting mart	Dashboards and analytics	ETL/ELT from transactional DB and events

13.2 Environment Strategy

Environment	Purpose	Notes
Local	Developer machine	Docker Compose or local services; fake SSO optional
DEV	Shared development integration	Feature branch deployments, test data
SIT	System integration testing	Integrate Core HR, Workflow, Recruitment, Onboarding
UAT	Business user testing	Masked realistic data, approval matrix testing
PRE-PROD	Production-like validation	Performance, DR, security, migration rehearsal
PROD	Live enterprise environment	HA, monitoring, backup, change control

13.3 Kubernetes / OpenShift Configuration Guidance

- Deploy each backend module as stateless pods with readiness and liveness probes.
- Use Horizontal Pod Autoscaler for API services based on CPU, memory, and request metrics.
- Use separate ConfigMaps for non-secret configuration and Secrets/Vault for passwords, tokens, signing keys, and database credentials.
- Use ingress/API gateway for TLS termination, routing, WAF policies, and rate limiting.
- Use network policies to restrict DB, broker, and storage access only to required namespaces/services.

- Use blue-green or canary deployment for performance and learning services to reduce production rollout risk.
- Use persistent storage only for databases, object storage, broker state, and reporting components; application pods should remain stateless.

13.4 Example Environment Variables

```
APP_ENV=prod
APP_MODULE=performance-service
DATABASE_URL=postgresql://<user>:<password>@<host>:5432/hrms
REDIS_URL=redis://<host>:6379/0
EVENT_BROKER_URL=<broker-connection-string>
OBJECT_STORAGE_BUCKET=hrms-documents
IAM_ISSUER_URL=https://sso.company.local/realms/hrms
IAM_AUDIENCE=hrms-api
AUDIT_ENABLED=true
WORKFLOW_SERVICE_URL=https://workflow-service.internal
NOTIFICATION_SERVICE_URL=https://notification-service.internal
LOG_LEVEL=INFO
FEATURE_AI_RECOMMENDATIONS=false
```

13.5 Backup and Disaster Recovery

Area	Requirement
Database	Daily full backup, point-in-time recovery, restore test at least quarterly
Object storage	Versioning, lifecycle policy, encryption, replicated backup if required
Event broker	Retention policy based on replay requirements; critical events stored in audit table
Configuration	Infrastructure-as-code and application configuration versioned in Git
RPO/RTO	Define based on HRMS criticality; suggested target: RPO <= 15 minutes, RTO <= 4 hours for production
DR drill	Run at least annually before performance cycle peak period

14. Reporting and Analytics

Dashboard / Report	Audience	Core Metrics
Performance Cycle Progress	HR, Managers	Eligible employees, self-review completion, manager review completion, overdue reviews
Rating Distribution	HR, Executives	Proposed rating, calibrated rating, final rating by department, grade, manager
Goal Completion	Managers, HR	Goal progress, overdue goals, achievement score
PIP Report	HR Employee Relations	Active PIPs, overdue milestones, closure outcome
Learning Compliance	HR Compliance, Executives	Mandatory assignment completion, overdue, waiver, expired certifications
Learning Effectiveness	L&D, HR	Completion rate, assessment score, training feedback, post-training improvement
Certification Expiry	Employees, Managers, Compliance	Expiring soon, expired, renewed, missing evidence
Skills Gap Closure	HR Talent, Executives	Gaps identified, learning assigned, gaps closed, readiness improved
Talent Evidence	HR Talent	Performance rating, learning history, certificates, development progress

14.1 Analytics Data Mart Guidance

- Build reporting views or a reporting mart so dashboards do not overload transactional tables.
- Use slowly changing dimensions for employee department, position, grade, and manager history.
- Use snapshot fact tables for final performance outcomes and learning compliance at period close.
- Exclude confidential free-text comments from broad executive dashboards unless anonymized and authorized.
- Support export to PDF/Excel/CSV based on role permissions and audit all exports.

15. Testing and Quality Assurance

Test Type	What to Test	Examples
Unit tests	Business rules and validations	Goal weight totals, rating calculation, due date rules, certificate expiry
Integration tests	Cross-module flows	Offer accepted -> onboarding -> learning assignment -> probation goals
Workflow tests	Approval routing and maker-checker	Maker cannot approve own course, rating override requires HR Head
Security tests	RBAC and sensitive data	Employee cannot view calibration notes or team ratings
Performance tests	Peak cycle and learning assignment loads	10,000 review forms generated, mandatory training assigned to all employees
Data migration tests	Legacy data load	Courses, employees, certificates, performance history
UAT	Business end-to-end process	Annual review cycle, training request approval, certification renewal
Regression tests	Existing HRMS integration	Recruitment, onboarding, workforce, succession unaffected

15.1 Minimum UAT Scenarios

21. HR creates annual performance cycle, checker approves, cycle is published to eligible employees.
22. Employee creates goals, manager reviews and locks goals.
23. Employee submits self-review, manager submits rating, HR checker reviews, department calibrates, final rating is released.
24. Low performance result creates PIP and linked learning assignment.
25. L&D creates a course, checker approves, course is published to the catalog.
26. Compliance creates mandatory training rule, employees receive assignments, overdue notifications are sent.
27. Employee uploads certificate, L&D checker validates evidence, certificate appears in learning transcript.
28. Succession development plan creates learning assignment and completion updates successor readiness evidence.
29. Recruitment offer accepted triggers onboarding learning and probation goals for new employee.
30. Unauthorized users cannot access confidential reviews, calibration notes, or PIP details.

16. Migration and Master Data Preparation

Before building or deploying the modules, the project team must prepare master data. Bad master data will break eligibility, approvals, reporting lines, assignment rules, and succession integration.

Data Area	Preparation Required
Employee master	Active employees, employment type, grade, department, manager, position assignment
Position and job data	Position ID, job ID, job family, grade, department, critical role flag if applicable
Competency framework	Competency library, proficiency levels, mapping to job/position/grade
Performance templates	Cycle types, review sections, rating scale, weighting rules
Learning catalog	Courses, paths, delivery mode, providers, prerequisites, owners
Mandatory training rules	Target population, due date, renewal frequency, escalation owner
Certification data	Certification types, issuers, expiry rules, evidence requirements
Workflow matrix	Approver roles, thresholds, SoD rules, SLA, escalation path
Historical data	Previous ratings, completed training, certificates, learning transcript if available

16.1 Data Migration Steps

31. Extract legacy performance and learning data into staging files/tables.
32. Normalize employee IDs and map legacy employee numbers to Core HR employee_id.
33. Cleanse course names, certification names, and competency terms into standard master data.

34. Validate manager assignments and department mappings.
35. Load static master data first: competency library, rating scale, course catalog, certification types.
36. Load employee historical data: previous ratings, completions, certificates, transcripts.
37. Run reconciliation reports: counts, missing employees, duplicate courses, expired certificates, orphan records.
38. Obtain HR sign-off before enabling production workflows.

17. Implementation Roadmap

Phase	Scope	Key Deliverables
Phase 1 - Foundation	Core HR integration, RBAC, workflow hooks, audit, shared competency framework	Master data, API contracts, permission matrix
Phase 2 - Learning MVP	Course catalog, learning paths, onboarding learning, mandatory assignments, transcript	LMS screens, assignment engine, completion validation
Phase 3 - Performance MVP	Performance cycle, goals, self-review, manager review, final approval	Performance screens, review workflow, rating outcome
Phase 4 - Integration	Performance-to-learning gaps, learning-to-succession evidence, recruitment/onboarding integration	Events, data flows, reporting views
Phase 5 - Advanced Controls	Calibration, 360 feedback, PIP, external training budget, certification renewal	Advanced workflows, compliance reports
Phase 6 - Analytics	Executive dashboards, skills gap closure, learning effectiveness, talent evidence	Data mart, dashboards, export controls
Phase 7 - Future Ready	AI recommendations, skills inference, talent marketplace, xAPI/LRS support	AI roadmap, extensibility APIs

18. Future-Ready Enhancements

Capability	Description	Preparation Needed Now
Skills graph	Common skills model across recruitment, performance, learning, and succession	Use normalized skill IDs instead of free text only
AI goal recommendation	Suggest goals based on job, department, strategy, historical goals	Store goal library and structured job expectations
AI learning recommendation	Recommend courses based on gaps, career path, performance, and role	Capture skill gaps and course-skill mapping
Skills inference	Infer skills from training, projects, certifications, performance comments	Store structured evidence and metadata
Talent marketplace	Match employees to projects, roles, and internal gigs	Standardize skills, availability, mobility preferences
xAPI / LRS	Track learning outside the LMS	Use extensible learning activity model and event ingestion
External content providers	Integrate LinkedIn Learning, Coursera, Udemy, internal academies	Design provider abstraction and content metadata mapping
Digital credentials	Issue verifiable badges/certificates	Store certificate metadata and issuer trust model
Predictive readiness	Predict successor readiness and attrition risk	Maintain longitudinal performance and learning history

19. Delivery Checklist and Acceptance Criteria

19.1 Developer Delivery Checklist

- Database migrations created for performance and learning tables.
- API endpoints implemented with authentication, RBAC, validation, and audit.
- Workflow integration implemented for all maker-checker processes.
- Events published and consumed for recruitment, onboarding, performance, learning, and succession integration.
- UI screens implemented for employee, manager, HR, L&D, checker, and executive roles.
- Unit, integration, workflow, and security tests completed.
- Reporting views and dashboards created.

- Documentation for APIs, events, environment variables, and deployment updated.

19.2 Infrastructure Delivery Checklist

- Environment namespaces/projects created for DEV, SIT, UAT, PRE-PROD, and PROD.
- Secrets configured through approved secrets manager.
- Database schemas, backups, PITR, indexes, and monitoring enabled.
- Object storage configured for course content and certificate evidence with encryption.
- Event broker topics/queues configured with retention and replay policy.
- Observability configured: logs, metrics, tracing, alerts, dashboards.
- CI/CD pipelines configured with build, test, security scan, deployment, rollback.
- Blue-green or canary deployment strategy implemented for production rollout.
- DR and restore procedures tested.

19.3 Acceptance Criteria

Area	Acceptance Criteria
Performance cycle	HR can create, submit, approve, publish, and close a cycle with complete audit trail.
Goal management	Employee/manager can create and lock goals with configured weights and approval rules.
Review process	Self-review, manager review, calibration, final approval, and employee acknowledgment work end-to-end.
Learning catalog	L&D can create, approve, publish, search, and assign courses.
Mandatory learning	System can assign mandatory learning by rule and track overdue/compliance status.
Certification	Employee can upload certificate, checker validates, expiry reminders are generated.
Integration	New hire gets induction learning and probation goals; performance gap creates learning assignment; learning completion updates talent profile.
Security	Unauthorized access to confidential ratings, calibration notes, PIP records, and certificates is blocked.
Reporting	Dashboards show cycle progress, rating distribution, learning compliance, skills gap closure, and certification expiry.
Audit	All maker-checker and high-risk actions have immutable audit records.

20. References

The design is vendor-neutral, but the HR capability model is aligned with enterprise HCM patterns commonly seen in SAP SuccessFactors talent management, performance and goals, and learning capabilities.

Reference	How It Influenced This Design
SAP SuccessFactors Talent Management	Used as a reference pattern for integrated recruiting, onboarding, learning, performance, succession, and skills-based talent processes.
SAP SuccessFactors Performance & Goals	Used as a reference pattern for goal management, real-time feedback, skills-based assessments, and performance cycles.
SAP Continuous Performance Management	Used as a reference pattern for ongoing check-ins, activities, achievements, and 1:1 feedback connected to goals.
SAP SuccessFactors Learning / Corporate LMS	Used as a reference pattern for learning catalog, skills, compliance training, learning assignments, and development tracking.